EXP. NO: 02

DATE: 30/7/26

STUDY OF NETWORK CABLES

AIM:

Study of different types of Network Cables.

a) Understand different types of network Cable.

Different type of Cables used in network are:

1. Unshielded Twisted pair (UTP) cable.
2. Shielded Twisted pair (STP) Cable.
3. Coaxial Cable
4. Fibre optic cable.

Cable type:

UTP

Category	Maximum Data Transmission
Category 3	10 bps
Category 5	Upto 100 Mbps
Category 5e	1 Gbps

Advantages:

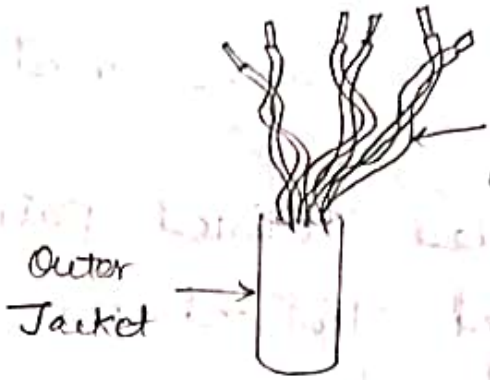
* Cheaper in cost
* Easy to install as they have a smaller overall diameter.

Disadvantages:

* More prone to (EMI) Electromagnetic interference and noise.

Application / Use:

* 10Base-T Ethernet
* Fast Ethernet, Gigabit Ethernet



Cable Type:

Cable Type	Category	Maximum Data Transmissi
STP	Category 6, 6a	10 Gbps
SSTP	Category 7	10 Gbps

Advantages:

* Shielded
* Faster than UTP
* Less Susceptible to noise and interference

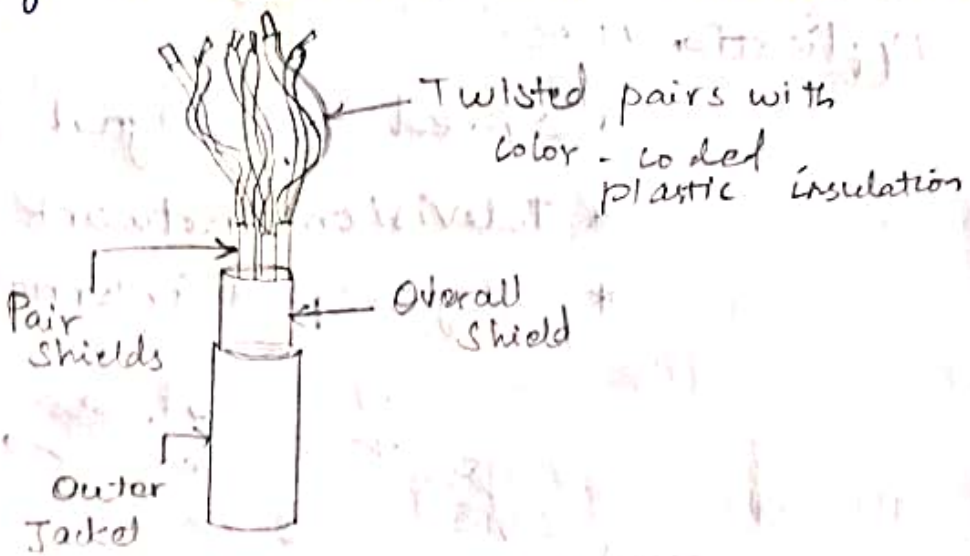
Disadvantages:

* Expensive
* Greater Installation Effort

Application / Use:

* Gigabit Ethernet, 10G Ethernet (55m) widely used in data centres
* Gigabit Ethernet, 10G Ethernet (100 m)

Cat



Cable Type:

Coaxial Cable

Category:

RG-6
RG-59
RG-11

Maximum data transmission

10 - 100 Mbps

Advantages:

* High bandwidth
* Immune to interference
* Low loss bandwidth
* Versatile.

Disadvantages:

* Limited distance
* Cost
* Size is bulky.

Application / Use:

* Speed of signal is 500 m.
* Television network
* High Speed internet connection.



Cable Type:

Fibre Optics Cable

Category:

Single Mode
Multi Mode

Maximum data Transmission:

100 Gbps

Advantages:

* High Speed
* High bandwidth
* High security
* Long distance

Disadvantages:

* Expensive
* Requires skilled installers.

Application / Use:

* Maximum distance of fibre optics cable is around 100 meters.



b) Make your own Ethernet Cross-Over Cable/ Straight Cable.

Tools and parts needed:

* Ethernet Cabling. CAT5e is Certified for gigabit support, but CAT5 Cabling works as well, just over Shorter distances.
* A crimping tool. This is an all-in one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the Cables.
* Two RJ45 plugs.
* Optional two plug shields.

RESULT:

Thus, all the cables are studied.